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B.Com. V Semester Degree Examination, May/June-2020
(CBCS Scheme 2016-17)
FUNDAMENTALS OF STATISTICS

Time : 3 Hours | Maximum Marks : 80

PART - A

Answer any **Five** Questions **2** marks each. (5×2=10)

1. Define Tabulation.
2. What is primary DATA?
3. Define Statistics.
4. Mention the Types of classification.
5. What is skewness?
6. What are the measures of Dispersion?
7. Mention the Types of Diagrams

PART - B

Answer any **Three** questions **5** marks each. (3×5=15)

8. Explain the characteristics of statistics.
9. Write a Note on Secondary Data.
10. Find the Geometric mean from the following data.

    85 , 5.81, 7.008 , 385 , 0.008 , 78, 10.821

11. The marks scored by 50 students in an examination paper are given below.

    30, 45, 48, 55, 39, 25, 31, 12, 18, 21,
    54, 59, 51, 33, 43, 44, 10, 38, 19, 26,
    47, 35, 37, 41, 46, 33, 51, 37, 58, 58,
    17, 19, 23, 26, 29, 38, 57, 36, 35, 44,
    43, 27, 31, 43, 22, 31, 47, 34, 18, 15.

    Prepare a Frequency table taking a class interval of 10, calculate the Arithmetic mean,

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(2) Sub-divided — " —
(3) Multiple - — " —

12. Calculate Median from the following data

Marks above :	0	10	20	30	40	50	60	70	80
No of Students:	150	140	100	80	80	70	30	14	0

PART - C

Answer any **Three** questions **10** marks each. **(3×10=30)**

13. What are the measures of central tendency? Explain.

14. What is median ? How do you calculate median under different series.

15. Draw a Histogram and Locate mode graphically from the following data, verify the result by Formula.

Marks :	0-10	10-20	20-30	30-40	40-50
Frequency :	5	12	22	10	3

16. Calculate Quartile deviation and its co-efficient from the data given below.

Wages :	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No of persons :	01	03	11	21	43	32	09

17. From the data given below calculate Bowley's Co-efficient of Skewness.

Age in years :	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
No of Persons :	5	7	8	18	15	12	7	5

PART - D

Answer any **One** question. **15** marks. **(1×15=15)**

18. Calculate mean, median and mode from the following data.

Marks :	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of Students :	5	10	16	30	21	10	8

19. Represent the following data

Items of Expenditure	Mo
Food	
Clothing	
Education	
House Rent	
Miscellaneous	
Total	

Answer

20. a) Pr ch

b)

19. Represent the following data by sub-divided bar diagram drawn on percentage basis.

Items of Expenditure	Monthly Expenditure in their families (in Rs)		
	Family - A	Family - B	Family - C
Food	200	280	600
Clothing	130	140	150
Education	150	140	100
House Rent	100	100	90
Miscellaneous	20	40	60
Total	**600**	**700**	**1000**

PART - E

(1×10=10)

Answer the following Question **10** marks.

20. a) Prepare a Questionnaire to Study the consumer satisfaction of any product of your choice.

(OR)

b) Prepare a percentage bar diagram with imaginary figures of production, sales and profit for 4 years of any organization.

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Fifth Semester B.Com. Degree Examination, November/December 2017
(Semester Scheme)
FUNDAMENTALS OF STATISTICS

Time : 3 Hours Max. Marks : 80

PART – A

ವಿಭಾಗ – ಎ

Answer **any five** questions. 2 marks **each**. (5×2=10)

ಯಾವುದಾದರೂ **ಐದು** ಪ್ರಶ್ನೆಗಳಿಗೆ ಉತ್ತರಿಸಿ. ತಲಾ **2** ಅಂಕಗಳು.

1. Mention two essentials of good measures of variation.
   ವಿಚಲತೆಯ ಎರಡು ಅವಶ್ಯಕತೆಗಳನ್ನು ತಿಳಿಸಿ.
2. State any two differences between Dispersion and Skewness.
   ವಿಚಲತೆ ಮತ್ತು ವಿಷಮತೆಯ ಎರಡು ವ್ಯತ್ಯಾಸಗಳನ್ನು ತಿಳಿಸಿ.
3. What are the types of Ogive curves ?
   ಓಜಿವ್ ರೇಖೆಯ ಪ್ರಕಾರಗಳಾವುವು ?
4. State the types of skewness.
   ವಿಷಮತೆಯ ವಿಧಗಳನ್ನು ತಿಳಿಸಿ.
5. Write the different parts of a table.
   ಪಟ್ಟಿಯ ವಿವಿಧ ಭಾಗಗಳನ್ನು ಹೆಸರಿಸಿ.
6. What is Geometric Mean ?
   ಗುಣೋತ್ತರ ಸರಾಸರಿ ಎಂದರೇನು ?
7. Define the term coefficient of variation.
   ವಿಚಲತೆಯ ಗುಣಾಂಕ ಎಂದರೇನು ?

P.T.O.

PART – B

ವಿಭಾಗ – ಬಿ

(4×10=40)

Answer **any four** questions. **10** marks **each**.

ಯಾವುದಾದರೂ **ನಾಲ್ಕು** ಪ್ರಶ್ನೆಗಳಿಗೆ ಉತ್ತರಿಸಿ. **ತಲಾ 10** ಅಂಕಗಳು.

8. Define statistics. Explain the functions and limitations of statistics.

ಸಂಖ್ಯಾಶಾಸ್ತ್ರವನ್ನು ವ್ಯಾಖ್ಯಾನಿಸಿ. ಸಂಖ್ಯಾಶಾಸ್ತ್ರದ ಕಾರ್ಯ ಮತ್ತು ಮಿತಿಗಳನ್ನು ವಿವರಿಸಿ.

9. What is Primary and Secondary data ? What are the methods of collecting Primary data ?

ಪ್ರಾಥಮಿಕ ಮತ್ತು ದ್ವಿತೀಯ ನ್ಯಾಸ ಎಂದರೇನು ? ಪ್ರಾಥಮಿಕ ನ್ಯಾಸವನ್ನು ಸಂಗ್ರಹಿಸುವ ವಿವಿಧ ವಿಧಾನವನ್ನು ವಿವರಿಸಿ.

10. a) A frequency distribution gives the following information, find the value of skewness.

ಈ ಕೆಳಗಿನ ಮಾಹಿತಿಯಿಂದ ವಿಷಮತೆಯನ್ನು ಕಂಡುಹಿಡಿಯಿರಿ.

i) Sum of the two quartiles = 22

ii) Difference between the two quartiles = 10

iii) Median = 11.

b) From the following data, find the mean and mode of the distribution.

ಕೆಳಗಿನ ನ್ಯಾಸದಿಂದ, ಸರಾಸರಿ ಮತ್ತು ಬಹುಮತ ಸರಾಸರಿಯನ್ನು ಕಂಡುಹಿಡಿಯಿರಿ.

I) Coefficient of variation = 5

II) Standard deviation = 2

III) Karl Pearson's coefficient of skewness = 0.5.

11. Calculate Bowley's co-efficient of skewness from the following data :

ಕೆಳಗಿನ ವಿವರಗಳಿಂದ ಬೌಲೆರವರ ವಿಷಮತೆಯ ಗುಣಾಂಕವನ್ನು ಕಂಡುಹಿಡಿಯಿರಿ.

Mid value	125	130	135	140	145	150	155	160	165	170	175
Frequency	12	25	32	44	36	50	38	20	15	20	13

12. Calculate geometric mean and Harmonic mean from the following data.

ಕೆಳಗಿನ ನ್ಯಾಸದಿಂದ ಗುಣೋತ್ತರ ಮತ್ತು ಸಾಮರಸ ಸರಾಸರಿ ಕಂಡುಹಿಡಿಯಿರಿ.

Values	3842	382	63	8	0.4	0.03	0.009	0.0005

13. Draw up a blank table to show the distribution of students of a college according to

ಕೆಳಗಿನ ವಿವರಗಳಿಗೆ ಖಾಲಿ ಪಟ್ಟಿಯನ್ನು ತಯಾರಿಸಿ.

a) Sex : Boys and Girls.
b) Age groups : Below 18 and above 18.
c) Class : PUC and Degree.
d) Faculty : Arts, Commerce and Science.
e) Period : 2016 and 2017.

PART – C

ವಿಭಾಗ – ಸಿ

(2×15=30)

Answer **any two** questions. **15** marks **each**.
ಯಾವುದಾದರೂ **ಎರಡು** ಪ್ರಶ್ನೆಗಳಿಗೆ ಉತ್ತರಿಸಿ. **ತಲಾ 15** ಅಂಕಗಳು.

14. Compute mean, median and mode from the following data :

ಕೆಳಗಿನ ಮಾಹಿತಿಯಿಂದ ಸರಾಸರಿ, ಮಧ್ಯಕ ಮತ್ತು ಬಹುಮತ ಸರಾಸರಿ ಕಂಡುಹಿಡಿಯಿರಿ.

Marks	0 – 9	10 –19	20 – 29	30 – 39	40 – 49	50 – 59	60 – 69	70 – 79	80 – 89	90 – 99
No. of Students	7	9	18	36	54	75	51	27	15	8

15. Calculate coefficient of variation and skewness based on mean, mode and standard deviation from the data given below :

ಸರಾಸರಿ, ಬಹುಮತ ಸರಾಸರಿ ಮತ್ತು ನಿಯತ ವಿಚಲನೆಯನ್ನು ಬಳಸಿ, ವಿಚಲನೆಯ ಗುಣಾಂಕ ಮತ್ತು ವಿಷಮತೆಯನ್ನು ಕಂಡುಹಿಡಿಯಿರಿ.

Monthly income (Rs.)	Below 100	100 – 150	150 – 200	200 – 250	250 – 300	300 – 350
No. of Workers	10	35	180	400	470	500

16. From the following data, find out :
    a) Which bags has the highest average bursting pressure ?
    b) Which has more uniform pressure ?

    ಕೆಳಗಿನ ನ್ಯಾಸದಿಂದ :
    1) ಯಾವ ಚೀಲಗಳು ಹೆಚ್ಚಿನ ಒತ್ತಡದಲ್ಲಿ ಹೊಡೆಯುತ್ತವೆ ?
    2) ಯಾವ ಚೀಲಗಳು ಹೆಚ್ಚಿನ ಏಕತೆ ಒತ್ತಡ ಹೊಂದಿವೆ ?

Bursting Pressure (lbs)	Numbers of Bags	
	Dhanush	Suhas
5.0 – 9.9	2	9
10.0 –14.9	9	11
15.0 –19.9	29	18
20.0 – 24.9	54	32
20.0 – 29.9	11	27
30.0 – 34.9	5	13
	110	110

17. Present the following data by percentage sub-divided bar diagram.
    ಕೆಳಗಿನ ನ್ಯಾಸವನ್ನು ಶೇಕಡಾವಾರು ವಿಭಜಿತ ಲಂಬ ರೇಖಾಂಶ ಚಿತ್ರದಲ್ಲಿ ಬರೆಯಿರಿ.

Particulars	2015	2016	2017
1. Cost per chair			
a. Wages	9	15	21
b. Other costs	6	10	14
c. Polishing	3	5	7
Total	18	30	42
2. Proceeds per chari	20	30	40
3. Profit (+)/Loss (–)	+2	–	–2

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V Semester B.Com. Degree Examination,
November/December 2019

(CBCS)

FUNDAMENTALS OF STATISTICS

Time : 3 Hours Max. Marks : 80

PART – A

Answer **any five** questions. **2** marks each : (5 × 2 = 10)

1. Define Statistics of Horace Secrist.
2. What is qualitative classification?
3. What is statistical investigation?
4. Define dispersion.
5. What is class frequency?
6. What is positive skewness?
7. What is histogram?

PART – B

Answer **any three** questions. **5** marks each. (3 × 5 = 15)

8. State any two differences between diagrams and graphs.
9. Mention any two differences between dispersion and skewness.
10. An average rainfall of Chitradurga city from Monday to Saturday is 0.3 inch. Due to heavy rainfall on Sunday, the average rainfall for the week increased to 0.5 inch. What was the rainfall on Sunday?
11. From moderately skewed distribution of retail prices for men's shoes, it is found that the mean price is Rs. 20 and median price is Rs. 17. If the C.V. is 20%, find Karl Pearson's coefficient of skewness.

12. From the following data find the value of the mean.

Central size :	10	12	14	16	18	20
Frequency :	3	7	12	20	8	5

PART – C

Answer **any** three questions. **10** marks each : **(3 × 10 = 30)**

13. What is mode? How do you calculate mode under different serics?

14. Briefly explain types of diagrams.

15. Tabulate the following information in one day trip organized by a college. There were 80 persons, each of whom paid Rs. 150 on an average. There were 60 students each of whom paid Rs. 120. Members of teaching staff were charged at a higher rate. The number of servants was 6 (all males) and they were not charged anything. The number of ladies was 20% of the total of which one was lady staff member.

16. Calculate geometric mean and harmonic mean for the following data :

Commodity :	H	K	S	G	B	C
Price relatives :	207	198	156	124	107	196

17. 20% of the workers in a firm employing a total of 2000 earned less than Rs. 2.00 per hour, 440 earned from Rs. 2.00 to 2.24 per hour, 24% earned from Rs. 2.25 to 2.49 per hour, 370 earned from Rs. 2.50 to 2.74 per hour, 12% earned from Rs. 2.75 to Rs. 2.99 per hour and the rest earned Rs. 3.00 or more per hour.

Set up frequency table and calculate model wage.

PART – D

Answer **any one** of the following. **15** marks each. **(1 × 15 = 15)**

18. Calculate coefficient of skewness based on quartiles from the following data :

Central value :	75	100	125	150	175	200	225	250
Frequency :	35	40	48	100	125	80	50	22

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(5x2=10)

The life of two types of tyres is given below. Find which of the two makes has higher average life? And which type would you prefer?

Life of Tyres in Kms	No. of Tyres	
	CEAT	MRF
5000-10000	18	15
10000-15000	22	14
15000-20000	26	30
20000-25000	25	28
25000-30000	09	13

PART – E

(1 × 10 = 10)

20. Answer the following, carries **10** marks :

(a) Prepare a questionnaire to study the consumer satisfaction of any product.

Or

(b) Prepare a percentage bar diagram (with imaginary figures) of production, sales and profit for 3 years of any organization.

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Fifth Semester B.Com. Degree Examination, November/December 2018
FUNDAMENTALS OF STATISTICS (New Syllabus)
(CBCS Scheme 2016-2017)

Time : 3 Hours

Max. Marks : 80

PART – A
ಭಾಗ - ಎ

Answer **any five** of the following questions. **2 marks each** : (5×2=10)
ಯಾವುದಾದರೂ **5** ಪ್ರಶ್ನೆಗಳಿಗೆ ಉತ್ತರಿಸಿ. ತಲಾ **2** ಅಂಕಗಳು :

1. What do you mean by statistical investigation ?
ಸಂಖ್ಯಾಶಾಸ್ತ್ರ ತನಿಖೆ ಎಂದರೇನು ?

2. What is secondary data ?
ಸಹಾಯಕ ಮಾಹಿತಿ ಎಂದರೇನು ?

3. What do you mean by dispersion ?
ವಿಚಲನೆ ಎಂದರೇನು ?

4. What are the types of averages ?
ಸರಾಸರಿಯ ವಿಧಗಳಾವುವು ?

5. What do you mean by Skewness ?
ವಿಷಮತೆ ಎಂದರೇನು ?

6. What is meant by Quartile Deviation ?
ಚತುರ್ಥ ವಿಚಲನೆ ಎಂದರೇನು ?

7. State any two limitations of Diagram.
ಚಿತ್ರಗಳಿಗಿರುವ ಯಾವುದಾದರೂ ಎರಡು ನ್ಯೂನತೆಗಳನ್ನು ತಿಳಿಸಿ.

PART – B
ಭಾಗ - ಬಿ

Answer **any three** of the following questions. **5 marks each** : (3x5=15)
ಯಾವುದಾದರೂ **3** ಪ್ರಶ್ನೆಗಳಿಗೆ ಉತ್ತರಿಸಿ. ತಲಾ **5** ಅಂಕಗಳು :

8. Explain the functions of Statistics.
ಸಂಖ್ಯಾಶಾಸ್ತ್ರದ ಕಾರ್ಯಗಳನ್ನು ವಿವರಿಸಿ.

P.T.O.

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10. Prepare a frequency distribution with exclusive class intervals (0-10, 10-20 so on) with tally bars.
(0-10, 10-20 ...)
76, 16, 24, 33, 85, 61, 79, 56, 28, 05, 86, 92, 44, 35, 66, 36, 13, 55, 28, 72.

11. Find Geometric mean :
2,574 475 75 05 0.8 0.08 0.005 0.0009

12. Mean = 45, Median = 48 and sk_p = –0.4, find Standard Deviation.
ಸರಾಸರಿ = 45, ಮಧ್ಯಗತ = 48 ಮತ್ತು sk_p = –0.4, ಪ್ರಮಾಣಿತ ವಿಚಲನ ಕಂಡುಹಿಡಿಯಿರಿ

PART – C
ಭಾಗ - ಸಿ

Answer **any three** of the following questions. **10** marks **each** : **(3×10=30)**
ಯಾವುದಾದರೂ **3** ಪ್ರಶ್ನೆಗಳಿಗೆ ಉತ್ತರಿಸಿ. ಪ್ರತಿ **10** ಅಂಕಗಳು :

13. What is tabulation and classification ? Explain parts of table.
ಸಾರಣಿಕ ಪಟ್ಟಿ ಮತ್ತು ವರ್ಗೀಕರಣ ಎಂದರೇನು ? ಸಾರಣಿಕ ಪಟ್ಟಿಯ ಭಾಗಗಳನ್ನು ವಿವರಿಸಿ.

14. What is primary data ? Explain various methods of collecting primary data.
ಪ್ರಾಥಮಿಕ ಮಾಹಿತಿ ಎಂದರೇನು ? ಅವನ್ನು ಸಂಗ್ರಹಿಸುವ ವಿವಿಧ ರೀತಿಗಳನ್ನು ವಿವರಿಸಿ.

15. Compute the coefficient of variation from the following data :
ಈ ಕೆಳಗಿನ ಮಾಹಿತಿಯಿಂದ ಗುಣಾಂಕ ವಿಚಲನೆಯನ್ನು ಕಂಡುಹಿಡಿಯಿರಿ.

Profit/Loss (in '000' Rs.) ಲಾಭ ಮತ್ತು ನಷ್ಟ (ರೂ. ಸಾವಿರಗಳಲ್ಲಿ)	No. of shops ಅಂಗಡಿಗಳ ಸಂಖ್ಯೆ
– 4 to –3	4
– 3 to – 2	10
– 2 to – 1	22
– 1 to – 0	28
0 to +1	38
+1 to +2	56
+2 to +3	40
+3 to +4	24
+4 to +5	18
+5 to +6	10

(iii) Which section is more consistent in scoring ?

ಯಾವ ವಿಭಾಗವು ಸ್ಥಿರವಾದ ಅಂಕಗಳನ್ನು ಪಡೆದಿದೆ ?

Marks / ಅಂಕಗಳು	0-10	10-20	20-30	30-40	40-50
No. of students / ವಿದ್ಯಾರ್ಥಿಗಳ ಸಂಖ್ಯೆ					
Boys Section / ಬಾಲಕರ ವಿಭಾಗ	9	13	20	10	8
Girls Section / ಬಾಲಕಿಯರ ವಿಭಾಗ	5	12	24	12	7

PART – E
ಭಾಗ - ಇ

(1×10=10)

Answer the following question. **10** marks **each** :

ಕೆಳಗಿನ ಪ್ರಶ್ನೆಗೆ ಉತ್ತರಿಸಿ. ತಲಾ **10** ಅಂಕಗಳು :

20. a) Prepare a questionnaire to study the consumer satisfaction of any product of your choice.

ನಿಮ್ಮ ಆಯ್ಕೆಯ ಯಾವುದೇ ಉತ್ಪನ್ನದ ಗ್ರಾಹಕ ತೃಪ್ತಿಯನ್ನು ಅಧ್ಯಯನ ಮಾಡಲು ಪ್ರಶ್ನಾವಳಿಯನ್ನು ತಯಾರಿಸಿ.

OR

b) Prepare a percentage bar diagram with imaginary figures of production, sales and profit for 3 years of any organisation.

ಯಾವುದಾದರೊಂದು ಸಂಘಟನೆಯ ಉತ್ಪಾದನೆ, ಮಾರಾಟ ಮತ್ತು ಲಾಭದ ಊಹಾ ಅಂಕಿ ಸಂಖ್ಯೆಗಳಿಂದ ಶೇಕಡಾವಾರು ವಿಭಜಿತ ಲಂಬರೇಖಾ ಮೂಲಕ ಚಿತ್ರವನ್ನು ಮೂರು ವರ್ಷಗಳಿಗೆ ತಯಾರಿಸಿ.

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B.Com. V Semester Degree Examination, May/June-2020
(CBCS Scheme 2016-17)
FUNDAMENTALS OF STATISTICS

Time : 3 Hours

Maximum Marks : 80

PART - A

Answer any **Five** Questions **2** marks each. (5×2=10)

1. Define Tabulation.
2. What is primary DATA?
3. Define Statistics.
4. Mention the Types of classification.
5. What is skewness?
6. What are the measures of Dispersion?
7. Mention the Types of Diagrams

PART - B

Answer any **Three** questions **5** marks each. (3×5=15)

8. Explain the characteristics of statistics.
9. Write a Note on Secondary Data.
10. Find the Geometric mean from the following data.
    85 , 5.81, 7.008 , 385 , 0.008 , 78, 10.821
11. The marks scored by 50 students in an examination paper are given below.
    30, 45, 48, 55, 39, 25, 31, 12, 18, 21,
    54, 59, 51, 33, 43, 44, 10, 38, 19, 26,
    47, 35, 37, 41, 46, 33, 51, 37, 58, 58,
    17, 19, 23, 26, 29, 38, 57, 36, 35, 44,
    43, 27, 31, 43, 22, 31, 47, 34, 18, 15.
    Prepare a Frequency table taking a class interval of 10, calculate the Arithmetic mean.

[P.T.O

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12. Calculate Median from the following data

Marks above :	0	10	20	30	40	50	60	70	80
No of Students:	150	140	100	80	80	70	30	14	0

PART - C

Answer any **Three** questions **10** marks each. **(3×10=30)**

13. What are the measures of central tendency? Explain.

14. What is median ? How do you calculate median under different series.

15. Draw a Histogram and Locate mode graphically from the following data, verify the result by Formula.

Marks :	0-10	10-20	20-30	30-40	40-50
Frequency :	5	12	22	10	3

16. Calculate Quartile deviation and its co-efficient from the data given below.

Wages :	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No of persons :	01	03	11	21	43	32	09

17. From the data given below calculate Bowley's Co-efficient of Skewness.

Age in years :	0-5	5-10	10-15	15-20	20-25	25-30	30-35	35-40
No of Persons :	5	7	8	18	15	12	7	5

PART - D

Answer any **One** question. **15** marks. **(1×15=15)**

18. Calculate mean, median and mode from the following data.

Marks :	10-20	20-30	30-40	40-50	50-60	60-70	70-80
No. of Students :	5	10	16	30	21	10	8

19. Represent the following data by sub-divided bar diagram drawn on percentage basis.

Items of Expenditure	Monthly Expenditure in their families (in Rs)		
	Family - A	Family - B	Family - C
Food	200	280	600
Clothing	130	140	150
Education	150	140	100
House Rent	100	100	90
Miscellaneous	20	40	60
Total	**600**	**700**	**1000**

PART - E

Answer the following Question **10** marks. (1×10=10)

20. a) Prepare a Questionnaire to Study the consumer satisfaction of any product of your choice.

(OR)

b) Prepare a percentage bar diagram with imaginary figures of production, sales and profit for 4 years of any organization.

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V Semester B.Com. Degree Examination, May/June 2014
(Semester Scheme)
BUSINESS STATISTICS - 1

Time : 3 Hours Max. Marks : 60

PART - A/ವಿಭಾಗ - ಎ

Answer any three questions, 5 marks each. (3×5=15)
ಯಾವುದಾದರೂ 3 ಪ್ರಶ್ನೆಗಳಿಗೆ ಉತ್ತರಿಸಿ. ಪ್ರತಿ 5 ಅಂಕಗಳು.

1. Define statistics according to Horace Secrist.
ಹೊರೇಸ್ ಸೆಕ್ರಿಸ್ಟ್‌ರವರ ಸಂಖ್ಯಾಶಾಸ್ತ್ರವನ್ನು ವ್ಯಾಖ್ಯಾನಿಸಿ.

2. State the advantages of diagrams.
ಚಿತ್ರಗಳಿಂದಾಗುವ ಅನುಕೂಲಕೆಗಳನ್ನು ತಿಳಿಸಿ.

3. An average rainfall of a city from Monday to Saturday is 0.3 inch. Due to heavy rainfall on Sunday, the average rainfall for the week increased to 0.5 inch. What was the rainfall on Sunday ?
ಸೋಮವಾರದಿಂದ ಶನಿವಾರದವರೆಗೆ ನಗರದ ಸರಾಸರಿ ಮಳೆಯ ಪ್ರಮಾಣ 0.3 ಇಂಚು. ಭಾನುವಾರ ಅತಿ ಹೆಚ್ಚು ಮಳೆಯ ಪ್ರಯುಕ್ತ ವಾರದ ಸರಾಸರಿ ಮಳೆಯ ಪ್ರಮಾಣ 0.5 ಇಂಚಿಗೆ ಜಾಸ್ತಿಯಾಗಿದೆ. ಭಾನುವಾರ ಮಳೆಯ ಪ್ರಮಾಣವೇನು ?

4. Locate median from the following :
ಕೆಳಗಿನ ದತ್ತಾಂಶಗಳಿಂದ ಮಧ್ಯಂಕ ಕಂಡುಹಿಡಿಯಿರಿ :

Size of shoe : ಪಾದರಕ್ಷೆ	5	5.5	6	6.5	7	7.5	8
Frequency : ಆವೃತ್ತಿ	10	16	28	15	30	40	34

P.T.O.

Given the data given below find mode:

Age	20-25	25-30	30-35	35-40	40-45	45-50	50-55	55-60
No. of Persons	50	70	80	180	150	100	70	50

PART – B/ಭಾಗ – ಬಿ

(2×10=20)

Answer any two questions. 10 marks each.

6. State the merits and demerits of mode.

7. Define dispersion. State the merits and demerits of standard deviation.

8. From the following information relating to 150 workers. Calculate average wages paid to workers.

Wages (₹) more than	75	85	95	105	115	125	135	145
No. of workers	150	140	115	95	70	60	40	25

9. Tabulate the following information in a trip organized by a college, there were 80 persons each of whom paid ₹ 115.50 on an average. There were 60 students each of whom paid ₹ 120. Members of teaching staff were charged at a higher rate. The no. of ladies was 20% of the total of which 1 was lady staff member. The no. of servants was 6 (all males) and they were not charged anything.

mean from the follow

, 5.81, 7.008, 385, 0.008, 78, 1

11. The marks scored by 50 students in an

PART – C/ಭಾಗ – ಸಿ

Answer any three questions. 15 marks each.
ಯಾವುದಾವುದೇ ಮೂರಕ್ಕೆ ಉತ್ತರಿಸಿ. ಪ್ರತಿ 15 ಅಂಕಗಳು. (3×15=45)

10. What is primary and secondary data ? Explain the methods of collecting primary data.

ಪ್ರಾಥಮಿಕ ಮತ್ತು ದ್ವಿತೀಯ ಸ್ಥಾನ ಎಂದರೇನು ? ಪ್ರಾಥಮಿಕ ಸ್ಥಾನವನ್ನು ಸಂಗ್ರಹಿಸುವ ವಿಧಾನಗಳನ್ನು ವಿವರಿಸಿ.

11. What are the different types of diagrams used in statistics ? Explain the merits and demerits of diagrams.

ಸಾಂಖ್ಯಿಕ ಅಂಕಿ ಅಂಶಗಳನ್ನು ಚಿತ್ರರೂಪದಲ್ಲಿ ನಿರೂಪಿಸುವ ವಿಧಾನಗಳಾವುವು ? ಸಾಂಖ್ಯಿಕ ಅಂಕಿ ಅಂಶಗಳನ್ನು ಚಿತ್ರರೂಪದಲ್ಲಿ ನಿರೂಪಿಸುವುದರ ಅನುಕೂಲ ಮತ್ತು ದುಷ್ಪರಿಣಾಮಗಳನ್ನು ವಿವರಿಸಿ.

12. Calculate Bowley's co-efficient of skewness from the following data :

ಕೆಳಗಿನ ಮಧ್ಯಾಂತರವಿಂದ ಬೌಲ್ಲರವರ ವಿಷಮತೆಯ ಗುಣಾಂಕವನ್ನು ಕಂಡುಹಿಡಿಯಿರಿ :

Mid value :	75	100	125	150	175	200	225	250
Frequency :	35	40	48	100	125	80	50	22

13. The goals scored by Pakistan and India team in the football matches were as follows. Find

ಈ ಕೆಳಗಿನ ಮಾಹಿತಿಗಳಿಂದ ಕೆಳಗಿನವುಗಳನ್ನು ಕಂಡುಹಿಡಿಯಿರಿ.

A) Which team is more goal taking ?

ಯಾವ ತಂಡವು ಹೆಚ್ಚು ಗೋಲುಗಳನ್ನು ಪಡೆದಿದೆ ?

B) Which team is more consistent ?

ಯಾವ ತಂಡವು ಹೆಚ್ಚು ಸ್ಥಿರತೆಯನ್ನು ಹೊಂದಿದೆ ?

No. of goals : ಗೋಲುಗಳ ಸಂಖ್ಯೆ	0	1	2	3	4
No. of matches : ಪಂದ್ಯಗಳ ಸಂಖ್ಯೆ					
Pakistan team : ಪಾಕಿಸ್ತಾನ ತಂಡ	24	9	8	5	4
Indian team : ಭಾರತ ತಂಡ	17	9	6	5	3

14. Present the following data by percentage sub-divided bar diagram.

ಈ ಅಂಕಿಗಳ ವ್ಯಾಪ್ತಿಯನ್ನು ಶೇಕಡಾವಾರು ವಿಭಜಿತ ಬಾರ ರೇಖಾಂಶ ಚಿತ್ರದಲ್ಲಿ ಬರೆಯಿರಿ.

Items ಮದ್ದುಗಳು	Family A ಕುಟುಂಬ ಎ (₹)	Family B ಕುಟುಂಬ ಬಿ (₹)
Food ಆಹಾರಧಾನ್ಯ	1,500	2,100
Clothing ಉಡುಪು	750	1,200
House rent ಮನೆಬಾಡಿಗೆ	600	900
Education ವಿದ್ಯಾಭ್ಯಾಸ	700	720
Fuel ಇಂಧನ	800	600
Others ಇತರೆ	650	480
Total ಒಟ್ಟು	5,000	6,000

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Fifth Semester B.Com. Degree Examination, May/June 2017
(2012 – 13 Syllabus) (New Scheme)
FUNDAMENTALS OF STATISTICS

Time : 3 Hours Max. Marks : 80

PART – A

Answer **any five** questions, **2** marks **each**. (5×2=10)

1. Define statistics according to Bowley.
2. What is primary data ?
3. What do you mean by class interval ?
4. What is diagram ?
5. Define median.
6. What is meant by quartiles ?
7. Define dispersion according to Spiegal.

PART – B

Answer **any four** questions, **10** marks **each**. (4×10=40)

8. Explain the differences between dispersion and skewness.
9. Explain the functions of statistics.
10. Present the following information in a suitable table supplying the figures not directly given.

    In 2006 out of 2000 workers in a factory, 1550 were members of a trade union. The number of women workers employed was 250, out of which 200 did not belong to any trade union.

    In 2007 the number of union workers was 1725 of which 1600 were men. The number of non-union workers was 380 among whom 155 were women.

11. From the following data calculate the missing value when its mean is 115.86

Wages (₹) :	110	112	113	117	?	125	128	130
No. of workers :	25	17	13	15	14	08	06	02

P.T.O.

12. Find out the mode of the following series.

Wages (₹)	No. of persons
Below 25	10
25 – 50	30
50 – 75	40
75 – 100	25
100 – 125	20
Above 125	15

13. Two workers on the same job show the following results over a long period of time.

Particulars	Worker A	Worker B
Mean time of completing the job (Minutes)	30	25
Std. deviation (minutes)	06	04

a) Which worker appears to be more consistent in the time to complete the job ?

b) Which worker appears to be faster in completing the job ?

PART – C

Answer **any two** questions, **15** marks **each**. **(2×15=30)**

14. Calculate Karl Pearson's coefficient of skewness from the following data.

Income in (₹) below :	200	300	400	500	600	700	800	900
No. of persons :	15	58	115	200	265	320	370	400

15. From the data given below calculate mean, median and mode.

Weight (kgs) :	10 – 19	20 – 29	30 – 39	40 – 49	50 – 59	60 – 69	70 – 79	80 – 89
No. of students :	08	32	81	152	92	24	09	02

16. A prospective buyer tests bursting pressures of samples of polythene bags received from two manufactures Kiran and Rajan. The test reveal the following results.

Bursting pressure :	10 – 15	15 – 20	20 – 25	25 – 30	30 – 35
No. of bags : Kiran :	3	14	42	82	9
Rajan :	13	22	45	48	22

Which manufactures bags have higher average bursting pressure ? Which of them is more uniform ?

17. Calculate Bowleys coefficient of skewness from the following data.

Marks	: 0 – 10	10 – 20	20 – 30	30 – 40	40 – 50	50 – 60
No. of stud.	: 12	18	30	20	15	5

ಕನ್ನಡ ಆವೃತ್ತಿ

ವಿಭಾಗ - ಎ

ಯಾವುದಾದರೂ ಐದು ಪ್ರಶ್ನೆಗಳಿಗೆ ಉತ್ತರಿಸಿ. ತಲಾ **2** ಅಂಕಗಳು. (5×2=10)

1. ಬೌಲಿಯವರ ಸಂಖ್ಯಾಶಾಸ್ತ್ರವನ್ನು ವ್ಯಾಖ್ಯಾನಿಸಿ.
2. ಪ್ರಾಥಮಿಕ ನ್ಯಾಸ ಎಂದರೇನು ?
3. ತರಗತಿ ಮಧ್ಯಾಂಕ ಎಂದರೇನು ?
4. ರೇಖಾ ಚಿತ್ರವೆಂದರೇನು ?
5. ಮಧ್ಯಾಂಕವನ್ನು ವ್ಯಾಖ್ಯಾನಿಸಿರಿ.
6. ಚತುರ್ಥ ಎಂದರೇನು ?
7. ಸ್ಪೈಗಲ್‌ರವರ ವಿಚಲನೆಯನ್ನು ವ್ಯಾಖ್ಯಾನಿಸಿ.

ವಿಭಾಗ - ಬಿ

ಯಾವುದಾದರೂ **ನಾಲ್ಕು** ಪ್ರಶ್ನೆಗಳಿಗೆ ಉತ್ತರಿಸಿ. ತಲಾ **10** ಅಂಕಗಳು. (4×10=40

8. ವಿಚಲನೆ ಮತ್ತು ವಿಷಮತೆಗಳಿಗಿರುವ ವ್ಯತ್ಯಾಸಗಳನ್ನು ವಿವರಿಸಿ.
9. ಸಂಖ್ಯಾಶಾಸ್ತ್ರದ ಕಾರ್ಯಗಳನ್ನು ವಿವರಿಸಿ.
10. ನೇರವಾಗಿ ಕೊಡದೆ ಇರುವ ಕೆಳಗಿನ ಮಾಹಿತಿಗಳಿಂದ ಪಟ್ಟಿಯನ್ನು ತಯಾರಿಸಿ. 2006 ನೇ ವರ್ಷದಲ್ಲಿ ಒಟ್ಟು 2000 ಕೆಲಸಗಾರರಿದ್ದು, 1550 ಕೆಲಸಗಾರರು ವ್ಯಾಪಾರ ಸಂಘಟನೆಯ ಸದಸ್ಯರಾಗಿರುತ್ತಾರೆ. ಮಹಿಳಾ ಕೆಲಸಗಾರರ ಸಂಖ್ಯೆ ಒಟ್ಟು 250, ಇದರಲ್ಲಿ 200 ಯಾವುದೇ ವ್ಯಾಪಾರ ಸಂಘಟನೆಯ ಸದಸ್ಯರಾಗಿಲ್ಲ.

    2007ರಲ್ಲಿ ವ್ಯಾಪಾರ ಸಂಘಟನೆಯ ಸದಸ್ಯರ ಸಂಖ್ಯೆ 1725 ಇದರಲ್ಲಿ 1600 ಕೆಲಸಗಾರರು ಗಂಡಸರಿದ್ದು 380 ಜನ ಕೆಲಸಗಾರರು ವ್ಯಾಪಾರ ಸಂಘಟನೆಯ ಸದಸ್ಯರಾಗಿಲ್ಲ, ಅದರಲ್ಲಿ 155 ಮಹಿಳಾ ಕೆಲಸಗಾರರಿದ್ದಾರೆ.
11. ಈ ಕೆಳಗಿನವುಗಳಿಂದ ತಪ್ಪಿಹೋದ ಕೂಲಿಯನ್ನು ಸರಾಸರಿ ಕೂಲಿ ರೂ. 115.86 ಇದ್ದಾಗ ಕಂಡುಹಿಡಿಯಿರಿ.

ಕೂಲಿ (ರೂ.)	:	110	112	113	117	?	125	128	130
ಕೆಲಸಗಾರರ ಸಂಖ್ಯೆ	:	25	17	13	15	14	08	06	02